

CS471 – Web Technologies (Laboratory)		Lab 1
		The Internet Protocols
Leen Suliman Odah	431201407	

Comparing TCP and UDP

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is a connection-oriented protocol that establishes a reliable connection using a 3-way handshake (SYN, SYN-ACK, ACK) before transmitting data. It guarantees packet delivery through acknowledgments (ACK) and retransmits lost packets. In contrast, UDP is connectionless and sends packets without setup or delivery confirmation.
Data Integrity and Ordering	TCP	TCP uses sequence numbers to ensure that data packets are reassembled in the correct order. It also uses checksums to detect corrupted data and retransmits lost packets. UDP does not use sequence numbers or guarantee packet ordering, delivery, or retransmission.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Applications that require high reliability and accurate data delivery where missing data is unacceptable (e.g., Web Browsing/HTTP/HTTPS, File Transfer/FTP, Email/SMTP, SSH)	Applications that require fast, real-time transmission and can tolerate minor packet loss (e.g., Video Streaming, Voice over IP/VoIP, Online Gaming, DNS, Live Broadcasting).
Performance	Slower with higher overhead due to dynamic windowing, flow control, congestion control, sequence tracking, and larger header size (20 bytes).	Faster and highly efficient with lower overhead due to minimal header size (8 bytes) and no connection management or delivery tracking.